Controller Design

Control Engineering

Course Schedule

1. Systems
2. Continuous-time Controllers
3. Laplace Domain
4. Poles/Zeros & Stability
5. Controller Design
6. Discrete-time Controllers
7. State Space Representation
8. Advanced Control Methods

Controller Design

- Feedback principles
- PID tuning
- Loop shaping
- Lead and Lag Compensators
- Root Locus
- Cascade control
- Ziegler–Nichols method
- T-Sum method

The Control Design Problem

- Given a plant
- Find a controller
- Meet stability, performance, and robustness requirements

Goals of Controller Design

- Stabilization: stabilize unstable or poorly damped plants
- Reference tracking: accurate and fast response to reference signals
- Disturbance rejection: attenuate disturbances and unknown inputs
- Robustness: tolerate modeling uncertainties while maintaining stability and acceptable performance

Feedback Control Principles

Error correction by controller:

1. measures output y
2. computes error e to reference (setpoint) r: $e = r - y$
3. acts to reduce it

→ Negative feedback

(positive feedback can increase gain and bandwidth but risks instability)

PID Tuning

Goal: determing controller parameters K_p, K_i, K_d to achieve desired closed-loop behavior

Typical tuning objectives: acceptable rise time, limited overshoot, small steady-state error, sufficient stability margin

Tuning approaches:

- model-based design (analytical)
- empirical tuning rules

Model-Based Control Design: Shape Loop Gain

For plant $G(s)$ and controller $C(s)$, loop gain $L(s) = C(s)G(s)$

needs to be known

Closed-loop transfer function $T(s) = \frac{L(s)}{1+L(s)}$

Stability depends on roots of $1 + L(s) = 0$

design $C(s)$ to achieve desired behavior (e.g., settling time or stability margins)

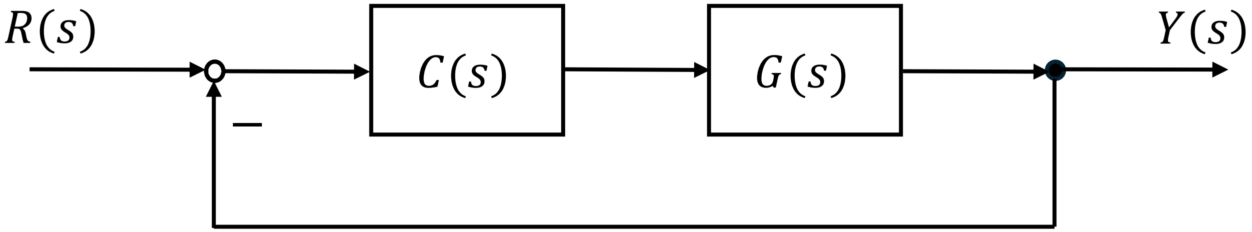
Key ideas:

- High loop gain at low frequency → good tracking & disturbance rejection
- Low loop gain at high frequency → noise attenuation & robustness

Setpoint Tracking

Response to reference changes
→ Transfer function from reference r to output y:

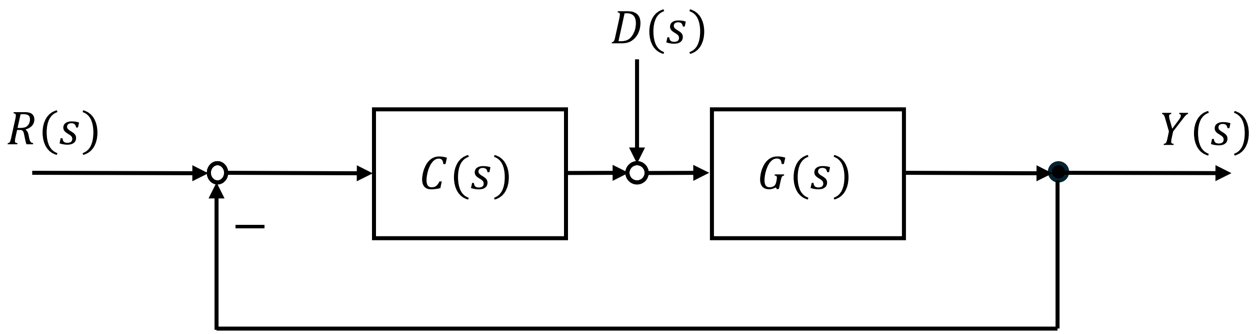
$$Y(s) = T(s) \cdot R(s)$$



Goals:
- Fast rise time
- Small overshoot
- Low steady-state error

→ High loop gain improves tracking accuracy: $T(s) = \frac{L(s)}{1+L(s)} \approx 1$

Disturbance Rejection



Response to external disturbances

If the disturbance enters at plant input:

$$Y(s) = T(s) \cdot R(s) + \frac{G(s)}{1+C(s) \cdot G(s)} \cdot D(s)$$

Disturbance sensitivity: $S(s) = \frac{1}{1+L(s)}$

$$Y(s) = T(s) \cdot R(s) + G(s) \cdot S(s) \cdot D(s)$$

→ High loop gain (small $S(s)$) improves disturbance rejection

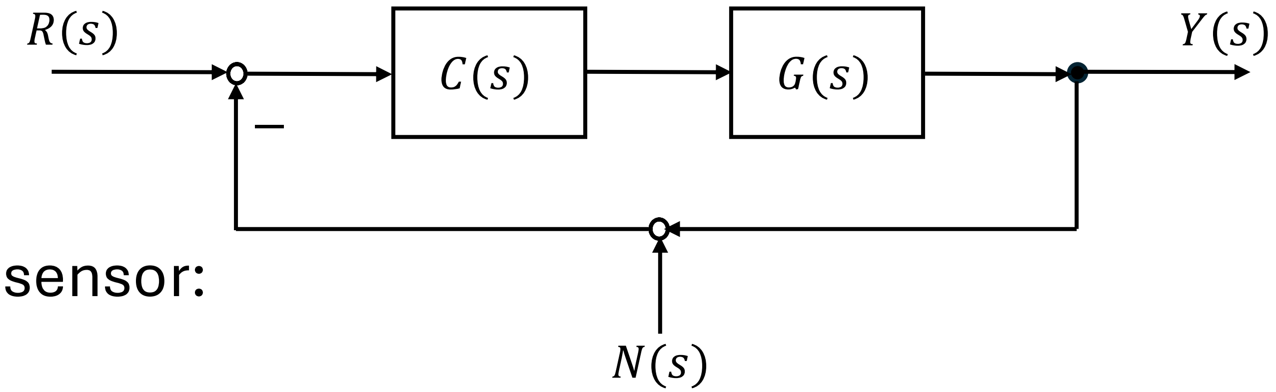
Sensitivity Functions

Measure effect of disturbances and uncertainties on closed-loop system

- Sensitivity function: $S(s) = \frac{1}{1+L(s)}$
- Complementary sensitivity function: $T(s) = \frac{L(s)}{1+L(s)}$

describe how signals propagate through the feedback loop

Measurement Noise



For measurement noise N added in the sensor:

$$Y(s) = T(s) \cdot R(s) - T(s) \cdot N(s)$$

Noise is interpreted by the controller as output error

Negative sign comes from feedback subtraction

→ Large T (high loop gain): noise amplification, small T: noise attenuation

But $T \approx 1$ required for good tracking → higher noise sensitivity

$S(s)$: how disturbances affect the output
$T(s)$: how reference and noise propagate

Trade-Off

$$S(s) + T(s) = 1$$

→ If S is small at some frequency, T must be near 1 (and vice versa)

→ Improving disturbance rejection inevitably affects noise sensitivity

Typical design goal:

- Low frequencies: make $|L|$ large → S small → good tracking and disturbance rejection
- High frequencies: make $|L|$ small → T small → suppress measurement noise

→ Controller $C(s)$ designed to shape loop gain $L(s) = C(s)G(s)$ to achieve this

Robustness

Describes how well a closed-loop system tolerates modeling uncertainties in the plant with respect to:

- robust stability (often assessed using stability margins)
- robust performance (maintaining acceptable tracking and disturbance rejection)

Performance-robustness trade-off:

- Aggressive control → fast tracking (and disturbance rejection) but reduced robustness (and higher noise sensitivity)
- Conservative control → more robust but slower tracking (and disturbance rejection)

→ High tracking and disturbance rejection typically require high loop gain, which can reduce robustness and increase noise sensitivity.

Common Model-Based Techniques

- Frequency response design (Bode plots)
- Root locus
- Internal Model Control (IMC) tuning
- Pole placement
- Optimization methods

Frequency Response Design

Bode plots (and sometimes Nyquist criterion as stability verification tool):

analyze the loop transfer $L(s)$ in the frequency domain

Design goal:

shape $|L(j\omega)|$ and $\angle L(j\omega)$ to meet stability and robustness margins

Example:

ensure $|L(j\omega)| \gg 1$ at low frequencies for tracking/disturbance rejection,
and $|L(j\omega)| \ll 1$ at high frequencies for noise suppression

Typical Loop Shaping Workflow

- Analyze plant frequency response $G(j\omega)$
- Choose desired crossover frequency (bandwidth)
- Adjust controller gain to set crossover
- Add phase lead to improve stability margins
- Add lag for improved low-frequency tracking
- Verify margins and time-domain response

Lead and Lag Compensators

Lead compensator:

- increases phase margin
- improves stability and transient response
- increases system bandwidth

Lag compensator:

- increases low-frequency gain
- improves steady-state tracking accuracy
- has little influence on phase margin

Root locus

Analyze poles of closed-loop system as a function of controller gain K

Root locus shows how loop gain $L(s) = K\ G(s)$ affects pole locations

→ Choose K (or $C(s)$) to place poles for desired transient behavior (settling time, damping ratio, stability margin)

(equivalent to shaping the low-frequency loop gain)

Interpretation:

- Poles moving toward the right half-plane → instability
- Poles further left → faster dynamics
- Complex poles with small damping → oscillations

IMC Tuning

Design $C(s)$ from a plant model so that closed-loop response behaves like a desired first-order/second-order system

Explicit loop shaping: set effective $L(s) = K\ G(s)$ to achieve target bandwidth and robustness

Advantages:

- systematic controller design based on plant model
- explicit trade-off between performance and robustness
- often leads to simple PI or PID structures

Limitations:

- requires reasonably accurate plant model
- performance degrades if model mismatch is large

Pole Placement

Select $C(s)$ so that closed-loop poles are at desired locations

Closed-loop poles are roots of $1 + L(s) = 0$

→ choosing pole locations directly shapes $L(s)$ in the s-plane

Typical steps:

1. Obtain plant model
2. Determine desired pole locations from performance requirements
3. Compute controller parameters so closed-loop poles match targets

- Advantage: direct control over system dynamics
- Limitations: requires accurate model and may require higher-order controller structures

Optimization Methods

Define a cost function (e.g., integral of squared error, sensitivity norms) and adjust paramters of $C(s)$ to minimize it

→ Implicitly shapes $L(s)$ to balance tracking, disturbance rejection, noise, and robustness

- Advantages: systematic and automated tuning, allows inclusion of constraints
- Limitations: requires numerical optimization, results depend on chosen cost function

Cascade Control

Concept:

- Two controllers arranged in nested loops
- Inner loop controls a fast subsystem
- Outer loop controls the main process variable

Advantages: improved disturbance rejection, faster response to internal disturbances, reduced effect of actuator nonlinearities

Typical applications: motor speed + torque control, temperature control with flow loop, robotics and process control

Empirical Approaches

- Model-based approaches: derive a system model, then use control design methods to compute parameters
- Empirical approaches: do experiments on the real plant, then use the measured response to compute parameters

Aspect	**Model-based**	**Empirical**
Knowledge of plant	Required	Not required
Method	Mathematical design	Experiment + rules
Tuning style	Analytical	Trial-based
Safety	Usually safer if model correct	May cause oscillations
Accuracy	Potentially higher	Moderate

Manual PI Tuning

Heuristic:

- Decide what response you want for a given error
- Compute the corresponding proportional gain K_p
- Choose the integral gain K_i as a fraction of K_p (e.g., $K_i = K_p/10$)

Example: Engine Torque of a Car Engine

- Cruise control deviation: $3\ \mathrm{m}/s$
- Desired torque response: $21\mathrm{Nm}$

→ $K_p = 7\ \mathrm{Nm}/(\mathrm{m/s})$

→ Integral gain ($1/10$ rule): $K_i = 0.7$

Ziegler–Nichols Method

Heuristic to provide practical starting values for P, PI, and PID controllers

Step-by-step procedure (closed-loop method)

1. Set controller to P-only mode: integral gain $K_i = 0$, derivative gain $K_d = 0$
2. Gradually increase proportional gain K_p
3. Find the point where the system output shows continuous sustained oscillations
4. Record ultimate gain K_u (gain at oscillation) and ultimate period T_u (oscillation period)
5. Use K_u and T_u to compute tuning parameters → empirical tuning formulas

Ziegler–Nichols Tuning Rules

$K_i = K_p/T_i$
$K_d = K_p/T_d$

Controller Type	K_p	T_i	T_d
P	$0.5\ K_u$	–	–
PI	$0.45\ K_u$	$T_u/1.2$	–
PID	$0.6\ K_u$	$T_u/2$	$T_u/8$

Advantages:

- Simple experimental procedure
- Widely used for initial PID tuning

Limitations:

- Can produce large overshoot
- Not ideal for highly nonlinear or unstable systems
- Often requires finetuning afterward

T-Sum (Time-Sum) Method

- Evaluate control system performance based on accumulated error over time
- Often used to compare different controller tunings (e.g., PID settings)

Key Idea: Measure how much control error accumulates over time

$$T_{\text{sum}} = \sum |e(t)| \cdot \Delta t$$

accumulated error over the evaluation period

control error

sampling interval

→ controller with smallest total accumulated error generally performs best

smaller T_{sum}: better control performance larger T_{sum}: larger or longer-lasting deviations

T-Sum Procedure

1. Apply a test input (often a step input)
2. Measure the system output response
3. Calculate the error signal:

reference signal

$$\underbrace{e(t) = r(t) - y(t)}$$

system output

4. Sum the error over the observation time

Limitations

- Does not distinguish between fast oscillations and slow drift
- Requires a defined evaluation time window

Exercise 1

Consider a unity feedback system with open-loop transfer function:

$$G(s) = \frac{10}{s+2}$$

- Write the closed-loop transfer function.
- Determine the steady-state gain.
- Explain how feedback changes system behavior.

Exercise 2

Given is a lead compensator:

$$C(s) = \frac{s+5}{s+20}$$

- Identify zero and pole.
- Explain its effect on phase and stability.

Exercise 3

Given is a lag compensator:

$$C(s) = \frac{s+1}{s+0.1}$$

Explain its effect and main purpose.

Exercise 4

Given:

$$G(s) = \frac{K}{s(s+2)}$$

- Identify poles and zeros.
- Describe root locus behavior as K increases.

Exercise 5

A system reaches sustained oscillations at:

- Ultimate gain: $K_u = 8$
- Oscillation period: $T_u = 2\ s$

Compute PID parameters using Ziegler–Nichols.